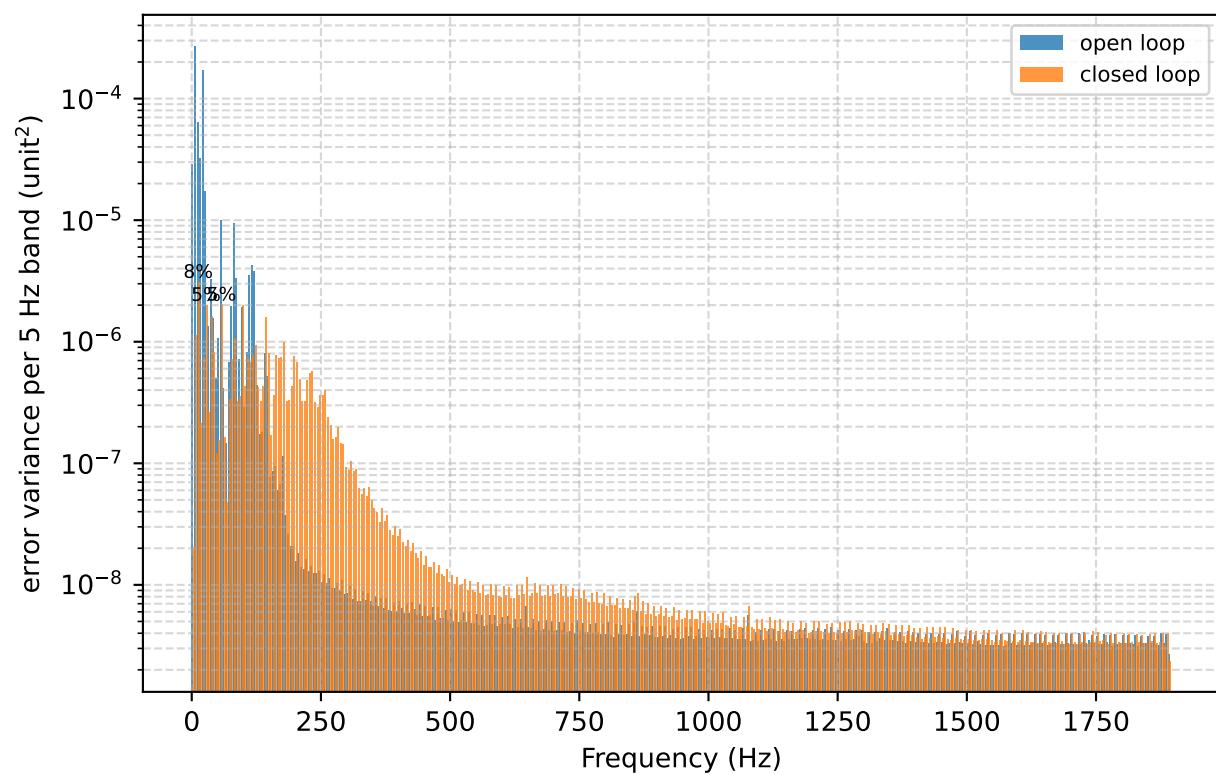
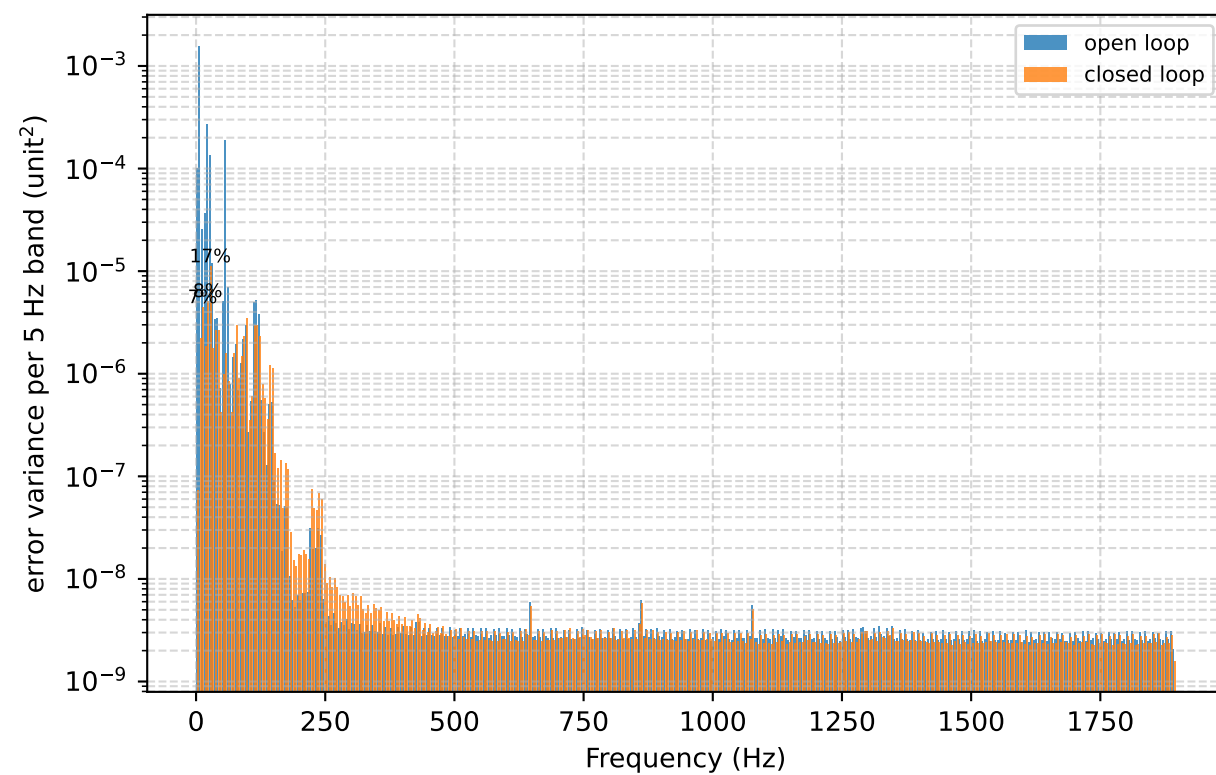
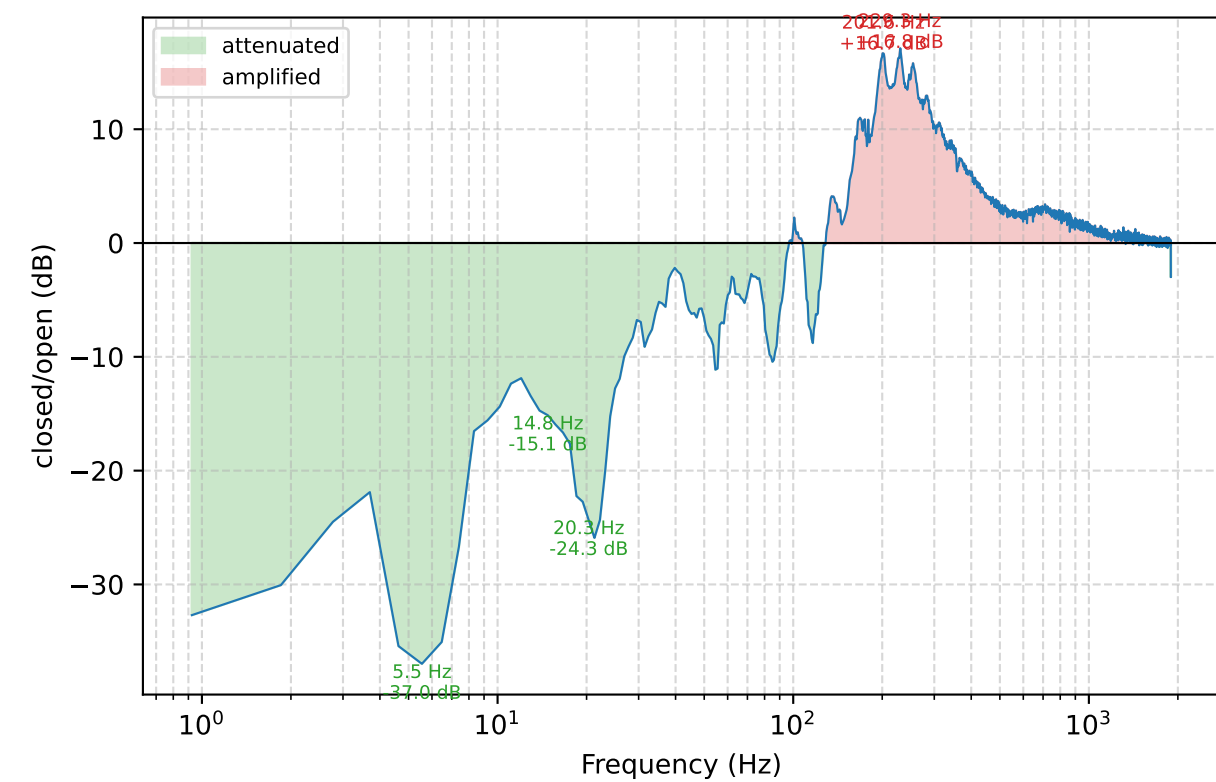
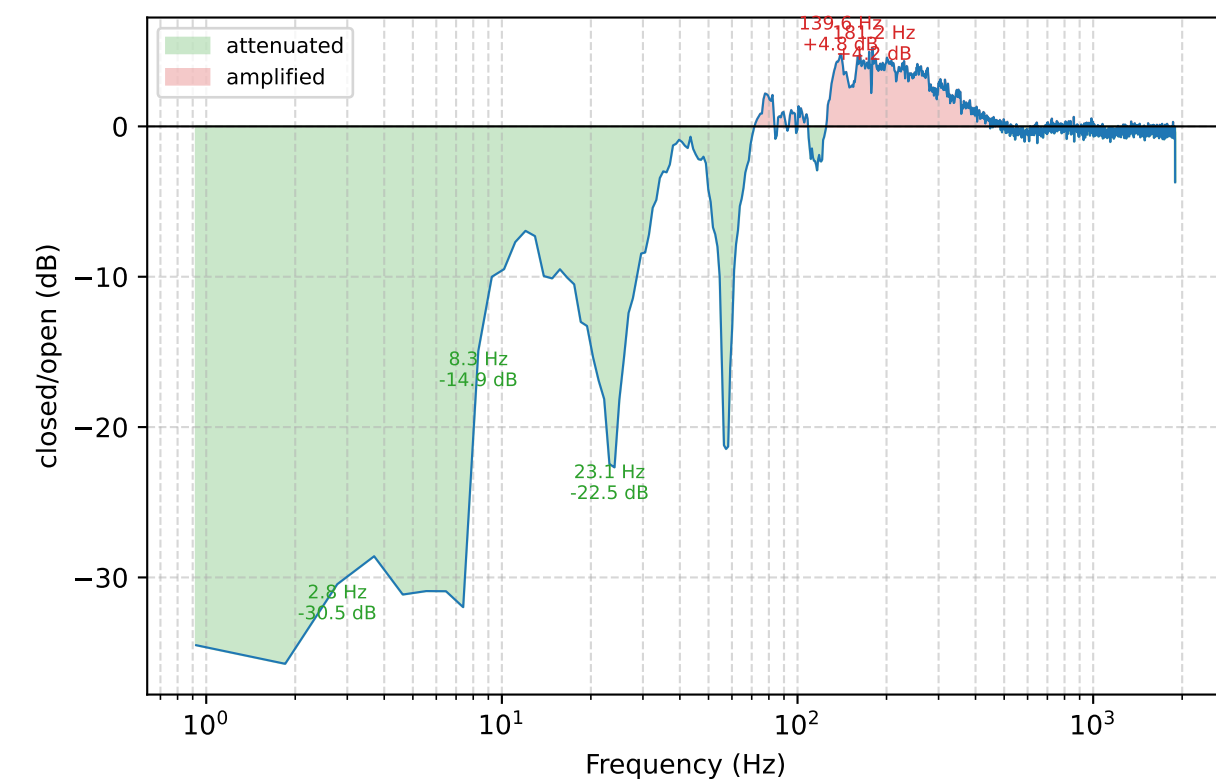
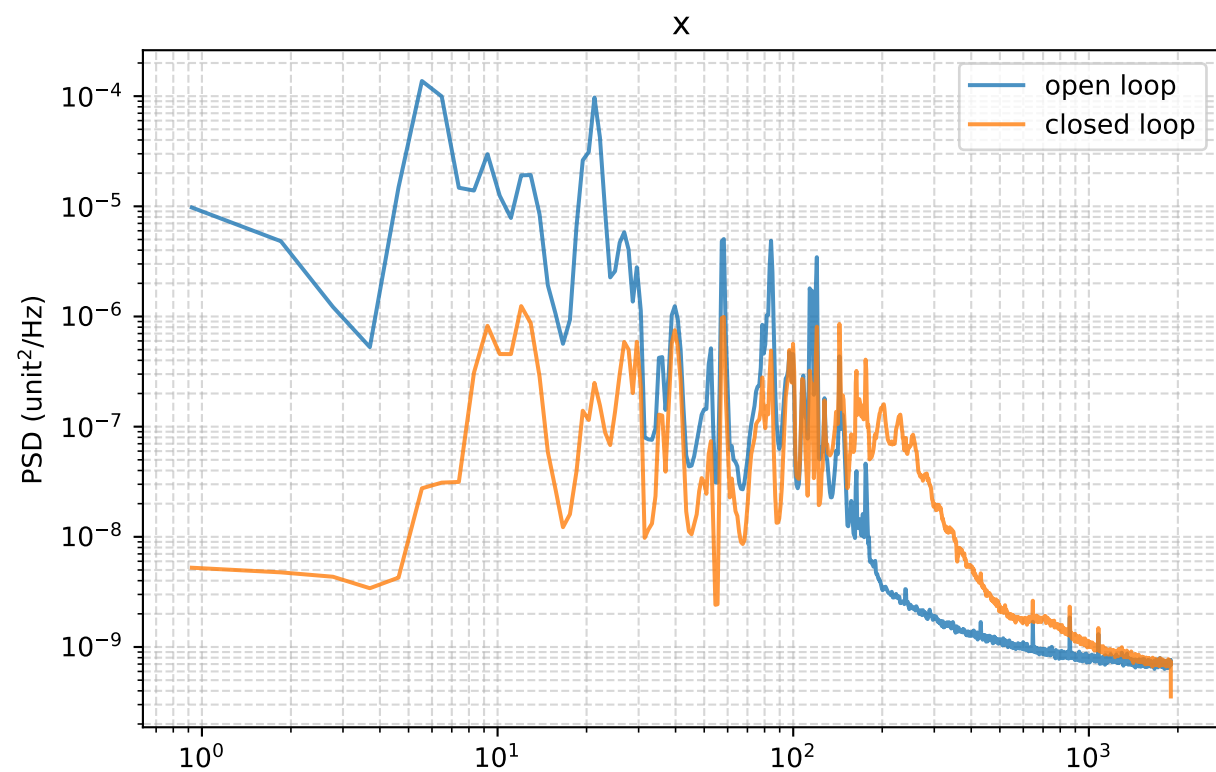
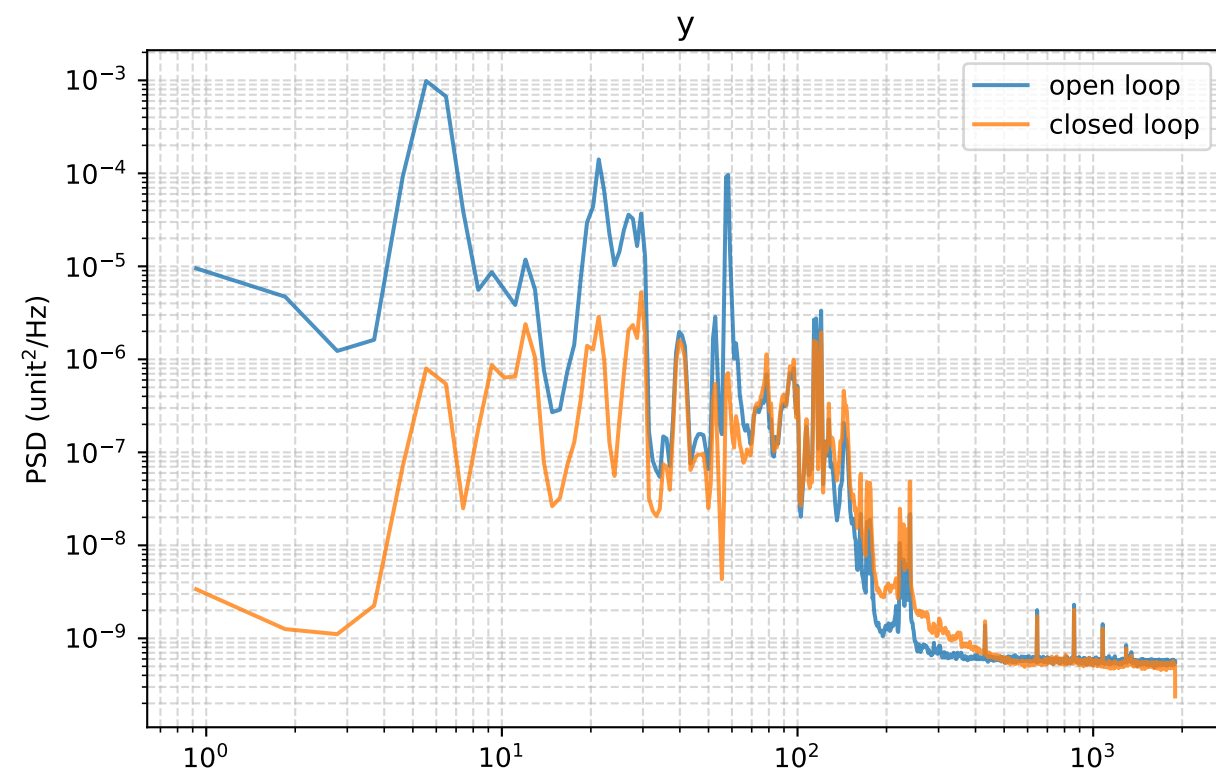


Closed-loop attenuation



Started: 2026-07-30 15:42:44
pixel_scale: 15.5 px per output unit
y: mean +1.834 -> -0.002 px; RMS about mean 0.946 -> 0.125 px (open -> closed)
x: mean +1.069 -> -0.003 px; RMS about mean 0.429 -> 0.097 px (open -> closed)

Config used:
ATTENUATION PAR-2 -- second closed-loop attenuation run through the PARPORT DAC (TI DAC81404 over the StarTech PEX1P2 / AX99100 PCIe parallel card, spi link, mmio backend), replacing the DAQC2/BRIDGEplate serial path used by every previous attenuation run.
TRANSPORT: bit-tilt 24-bit SPI, -6.4 us per channel update (-12.8 us for both channels), against the BRIDGEplate's -0.5-1.4 ms. DAC range is FIXED at +5 V bipolar (153 uV/LSB) with the mirror parked at 0 V, matching the bode sweeps exactly; auto-ranging is off, so the actuator is a fixed-gain element for both phases of the A/B and the open/closed comparison is clean. Supplies +/-12 V: AVDD needs VMAX+1.5 = 11.5 V for +10, AVSS -11.5 V, leaving only 0.5 V of margin -- meter the rails under load. AVSS is therefore NEGATIVE, not grounded, which means the 0-5 base range runs outside the AVSS = 0 V condition SLASEH2A 7.5 gives the unipolar figures at (see _avss_tradeoff); 16-bit = 76 uV/LSB in 0-5, 305 uV/LSB in +/-10, vs the BRIDGEplate's 12-bit 1 mV.
OUTPUTS: two channels, no coarse stage -- ch0 = y/tip (+1.0 V/unit), ch1 = x/tilt (-1.0 V/unit), offset 0 V, symmetric command bounds +1 V (59 px of authority against a +/-16 px ROI; the mirror's full range is deliberately NOT given to the loop on this first run). NOTE the command->FSM voltage mapping is NOT the same as any previous attenuation run, so K is not comparable across the transport change and was re-measured -- and it came out 4.7-5.5x HIGHER, not lower as this file originally predicted, so nothing about the piplate-era gains carries over.
Camera/CoM: 32x32 ROI start x 1840 / start y 680, exposure 50, gain 100, bw 100, 20x20 tracked window thr 5 (the camera is now shrouded, so the background floor no longer needs a high threshold).
FSP: broad_order 128, broad_mu .005 (the attenuation12 point; run 1 used the superseded attenuation11 512/.03), broad_lp 5 (retuned: the shorter delay moved the regeneration band from ~325 Hz to ~700 Hz); adapt_period 8, adapt_tau 5. Guard ON (guard_ratio 4). Cold-start observer: pass_open false + broad_freeze_closed false, so no weight is derived from open-loop data; drift_tau and the transient path are OFF. Modal freqs re-derived from run 1's own open PSD (8-mode cap), though they are inert while broad_order > 0.
RUN 1 RESULT TO BEAT (attenuation par fsp1, same config except 512/.03): y 0.518 -> 0.0084 px, x 0.496 -> 0.0966 px; 60-150 Hz only 1.5x and the largest closed contributor on both axes; waterbed 3.4-3.5x at 195-250 Hz; no regeneration (400-800 Hz = 0.0% of command power).
DELAY AND K: BOTH re-fit for this transport 2026-07-30 from conf_bode_par x/y.json (+/-100 mV sweep, full-band 5-300 Hz phase slope for tau, <=30 Hz plateau for K). Kx 3.787 +/- 0.008, Ky 3.636 +/- 0.005; delay x 2.717 fr, y 2.577 fr at fs 3788. Against the piplate era that is 4.7-5.5x MORE loop gain and LESS THAN HALF the delay -- the shorter delay is the entire reason to expect better rejection here.
COMPARE against attenuation_fsp21 and attenuation_fsp15 (run 15: y 0.073 / x 0.168 px): report total and per-band closed RMS, 60-400 Hz waterbed, command/clamp behavior, and transient activation count on the shared 500/500/4096 grid. Closed RMS in PIXELS remains comparable across the change; command units do not. refs: Kulcsar 2006 / Petit 2008 / Meimon 2010 / Correia 2010

RMS contributed by each 10 Hz band (px)				
band (Hz)	y open	y closed	x open	x closed
0-10	0.6355	0.0235	0.2692	0.0166
10-20	0.1226	0.0390	0.1522	0.0283
20-30	0.3136	0.0626	0.2124	0.0258
30-40	0.0610	0.0326	0.0334	0.0213
40-50	0.0317	0.0272	0.0223	0.0151
50-60	0.2162	0.0250	0.0515	0.0229
60-70	0.0433	0.0175	0.0116	0.0071
70-80	0.0286	0.0332	0.0251	0.0160
80-90	0.0216	0.0239	0.0556	0.0184
90-100	0.0353	0.0373	0.0253	0.0238
100-110	0.0139	0.0151	0.0163	0.0170
110-120	0.0497	0.0378	0.0432	0.0187
120-130	0.0323	0.0274	0.0319	0.0181
130-140	0.0098	0.0151	0.0092	0.0135
140-150	0.0158	0.0238	0.0179	0.0240
150-160	0.0059	0.0083	0.0063	0.0113
160-170	0.0041	0.0068	0.0061	0.0191
170-180	0.0049	0.0078	0.0066	0.0205
180-190	0.0020	0.0032	0.0039	0.0125
190-200	0.0017	0.0027	0.0032	0.0170
200-210	0.0018	0.0029	0.0029	0.0168
210-220	0.0019	0.0028	0.0026	0.0125
220-230	0.0035	0.0055	0.0026	0.0158
230-240	0.0035	0.0053	0.0025	0.0146
240-250	0.0028	0.0042	0.0025	0.0125
250-260	0.0014	0.0022	0.0023	0.0136
260-270	0.0014	0.0021	0.0023	0.0103
270-280	0.0013	0.0019	0.0021	0.0088
280-290	0.0013	0.0018	0.0022	0.0091
290-300	0.0013	0.0017	0.0022	0.0076
total 0-300	0.7598	0.1237	0.3915	0.0938